

image, with the highest option being whatever is the maximum resolution the sensor supports along with a few lower resolution options.

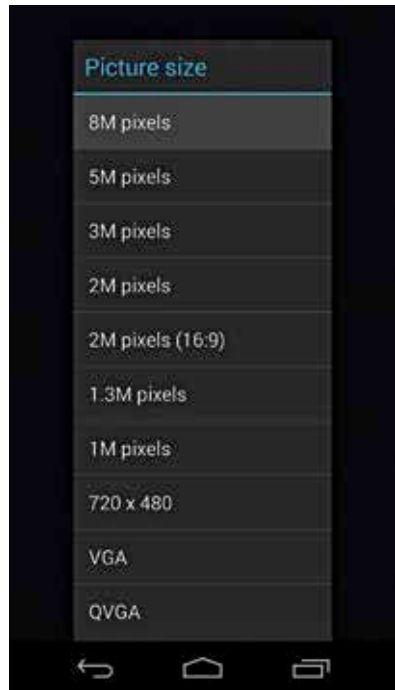
As some of you may know, resolution is measured in pixels. Because there is usually a lot of them, they are grouped together in a bunch of million pixels and called a megapixel. A common resolution these days is 8 megapixel, which means the sensor is capable of capturing 8 million pixels at once.

So how important is resolution and how important are megapixels? A lot and at the same time, not so much. A higher resolution image can have more detail, provided it is captured natively and not just an upscaled version of a low resolution image,

in which case the resolution will increase but the amount of detail will remain the same. So on a technical level, yes, an 8 megapixel image is better than a 5 megapixel image. But it's rarely that simple.

Often, as it happens, some sacrifices have to be made to achieve a higher resolution image. The ideal thing to do would be to keep the size of the pixels on the sensor the same and just add more pixels every time you wanted to increase its resolution. But this would also increase the physical size of the sensor, which is not ideal, especially on a smartphone, where space is at a premium. So often, the sensor size is kept mostly the same while increasing the resolution, which results in smaller pixels. Think of this as cutting a pizza into eight slices instead of six. It's the same size pizza so even though you are getting more slices, the individual slices are now smaller.

What happens when the pixels on the sensor are smaller? They are able to capture less light, which results in grainier images in low light situations



All those beautiful pixels